

Giacomo Cappelletto

Software Engineer and D1 Crew Athlete

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PROFESSIONAL SUMMARY

Dedicated Computer Engineering student with a strong academic record and hands-on experience in full-stack development, mobile application design, and machine learning. Proficient in modern frameworks (Next.js, React, Ruby on Rails) and experienced in cloud deployment, API integration, and Agile methodologies. Demonstrated ability to rapidly learn new technologies, optimize development processes, and deliver measurable improvements. Accomplished international rowing athlete competing for Boston University Men's Crew.

EDUCATION

- **Boston University, Boston, MA** 2024 – 2028
B.Sc. in Computer Engineering; GPA: 3.9
- **H-Farm International School, Venice, Italy** 2022 – 2024
International Baccalaureate Diploma Programme Higher Levels in Mathematics, Physics, and Computer Science

EXPERIENCE

- **TickIT GmbH, Boston, MA (Remote)** Oct 2024 – Jan 2025
Frontend & Backend Engineer – Ticketing Platform Startup
 - **Overview:** Engineered responsive user interfaces using Next.js, Shadcn components, and Tailwind CSS; developed robust backend functionality with Ruby on Rails and PostgreSQL, ensuring scalable API integrations.
 - **Contribution:** Provided key architectural insights on ticket authentication flow and streamlined Agile development processes while rapidly mastering new frameworks.
- **SOCIETA CAPPELLETTO S.R.L., Treviso, Italy** May 2024 – Jul 2024
Full Stack Native App Developer – Online Inventory Management System
 - **Project:** Developed an Electron-React-Node.js application integrating the Shopify API and Firebase Firestore to dynamically update online shop inventory across two locations.
 - **Impact:** Eliminated online inventory discrepancies and reduced manual labor times at checkout by 80%, significantly improving operational efficiency.
- **H-Farm International School, Treviso, Italy** Mar 2024 – Apr 2024
Python Developer – Student Meeting Application Developer
 - **Application:** Designed and implemented a Python Streamlit application to leverage school timetables from Managebac, automating meeting request emails using local LLMs in Ollama.
 - **Collaboration:** Collaborated closely with faculty to tailor the solution, ensuring seamless integration with existing systems and improved scheduling efficiency.
- **DLF Treviso, Treviso, Italy** Sep 2023 – Nov 2023
iOS Developer – Rowing LogBook Application
 - **Development:** Designed and developed an iOS app using Swift/SwiftUI (with Swift Charts) to enable athletes to log training sessions and analyze performance data.
 - **Process:** Managed full-cycle development from planning and design to implementation and evaluation, incorporating direct user feedback for continuous improvement.
- **E-Trash, Treviso, Italy** Feb 2023 – Jun 2023
CAD Engineer Intern
 - **CAD Modeling:** Designed and developed the first prototype of an intelligent trash bin using Autodesk Fusion 360, focusing on efficiency and structural integrity.
 - **Optimization:** Applied topology optimization and load testing to enhance the prototype's durability and weight distribution.
 - **3D Rendering:** Created high-quality 3D visualizations in Blender for presentation and product refinement.

PROJECTS

- **NoteWorthy – Automated Note Conversion Platform:** Led development of noteworthyxyx a Next.js/TypeScript application that converts handwritten notes into styled PDFs and LaTeX code. Integrated the Google AI Studio API for content processing, implemented GitHub/Google OAuth for secure authentication, containerized a Node.js LaTeX compiler with Docker, and deployed via Google Cloud Run on Vercel.
- **CIFAR-10 Image Recognition with Neural Architecture Search:** Engineered a machine learning solution using TensorFlow/Keras to perform image recognition on the CIFAR-10 dataset. Developed a neural architecture search pipeline that generated and evaluated CNN architectures with convolutional layers, max-pooling, and global average pooling. Automated candidate training, performance logging, and selection of the best-performing model for extended training and deployment.

SKILLS

- **Languages:** JavaScript (JSX/TSX, JS), Python, Ruby, C/C++, Swift/SwiftUI, Java
- **Frameworks & Libraries:** Next.js, React.js, Ruby on Rails, Node.js, Electron, SwiftUI
- **Technologies:** Tailwind CSS, PostgreSQL, Docker, Firebase, Shopify API, Streamlit, Ollama, Google Cloud Run, Git, AutoCAD, Autodesk Fusion 360
- **Methodologies:** Agile Development, API Integration, Cloud Deployment, Topology Optimization, Load Testing, 3D Rendering